

Quadratic functions summary

Shape → turning point → completing the square

01 Start with the shape

Look at a , the coefficient of x^2 in $y = ax^2 + bx + c$.

$$a > 0$$

SMILE = minimum

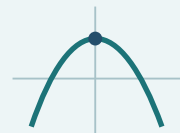
Lowest turning point



$$a < 0$$

FROWN = maximum

Highest turning point



02 Read the turning point

$$y = a(x - h)^2 + k$$

turning point = (h, k)

INSIDE THE BRACKET

Reverse its sign for x .

CONSTANT OUTSIDE

Keep its sign for y .

Example: $y = 2(x - 4)^2 - 7$ → turning point $(4, -7)$; minimum = -7 .

Tip: $x - 4 = 0$ gives $x = 4$; the outside constant is -7 , not $+7$.

03 Complete the square

For a leading coefficient of 1, halve the x -coefficient and square it:

$$x^2 + ax + b = (x + a/2)^2 - (a/2)^2 + b$$

The $+$ and $-$ of $(a/2)^2$ cancel, so the expression is unchanged.

Here a is the x -coefficient, not the x^2 -coefficient from Section 01.

If x^2 has a coefficient other than 1, factor it out of the x terms first.

$$f(x) = 2x^2 - 8x + 5$$

$$= 2(x^2 - 4x) + 5$$

$$= 2[(x - 2)^2 - 4] + 5$$

$$= 2(x - 2)^2 - 3$$

READ OFF

$$\text{TP} = (2, -3)$$

minimum = -3

range: $y \geq -3$

Check: a square is never negative, so $2(x - 2)^2 - 3$ cannot go below -3 .

Quadratic functions summary

Solve an equation → check its sign → apply it

04 Solve for x using completed-square form

Make the square stand alone, then take BOTH square-root signs.

Solve: $x^2 - 6x + 5 = 0$

Complete the square: $(x - 3)^2 - 9 + 5 = 0$

Simplify: $(x - 3)^2 - 4 = 0$

Isolate the square: $(x - 3)^2 = 4$

Square root: $x - 3 = \pm 2$

So $x = 3 + 2 = 5$ OR $x = 3 - 2 = 1$.

Remember $\pm$: a positive square can come from a positive or negative value.

If the isolated square equals a negative number, there are no real solutions.

05 Always positive or always negative?

In $y = a(x - h)^2 + k$, check the minimum or maximum value k.

ALWAYS POSITIVE

$a > 0$ and $k > 0$

ALWAYS NEGATIVE

$a < 0$ and $k < 0$

If $k = 0$, the graph touches $y = 0$: it is not strictly positive or negative.

06 Use it as a model

A rectangle has sides x cm and $(12 - x)$ cm. What is its greatest area?

$A(x) = x(12 - x) = -(x - 6)^2 + 36$

$0 < x < 12$; maximum area = 36 cm^2 when $x = 6 \text{ cm}$. Check domain and units.